

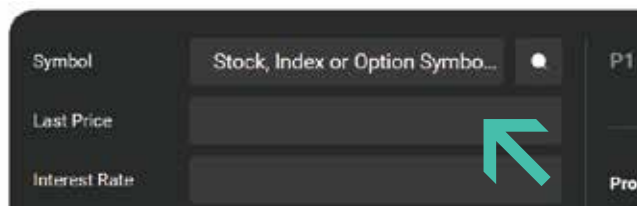
Probability Calculator

Powered by IVolatility

The Probability Calculator, powered by IVolatility.com allows you to enter a stock, index, or ETF to populate information on its potential price movement. The calculator will provide you with the last price of the instrument, current dividend and interest rate information, and allow you to select a target expiration date. Using this information and custom price inputs, the calculator provides a statistical probability of the underlying touching either of two prices (P1 or P2) or settling in between the two and touching neither. Note that all data is provided with a 20-minute delay. If the markets are closed, the previous trading day's market closing price will be used.

Choose a Stock Symbol, Index Symbol, or Option Symbol

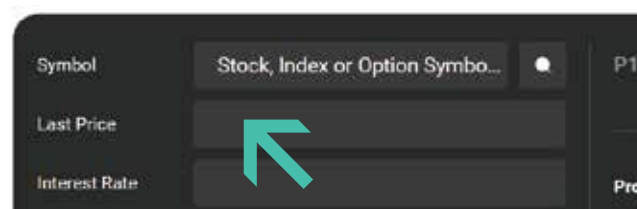
Input any stock, index or option symbol. If you don't know the symbol, you can browse by company name, instrument, market or exchange via the "Symbol Lookup" icon. Once you select a symbol, data will populate the table with the underlying's current price, interest rate and dividend data. Users can adjust all of these fields.



This screenshot shows the top section of the Probability Calculator interface. It includes a 'Symbol' field with a dropdown menu labeled 'Stock, Index or Option Symbo...'. To the right of the dropdown is a small circular icon with a dot. Below the 'Symbol' field are two more input fields: 'Last Price' and 'Interest Rate'. A teal arrow points from the 'Last Price' field towards the 'Interest Rate' field. On the far right, there are labels for 'P1' and 'Prob'.

Last Price

The option's underlying price is 20-minute delayed. If the markets are closed, the previous trading day's market closing price will be used. The user can change it to today's current price or any other price by manually entering the desired price. Notice that the price must be entered in decimal format.



This screenshot shows the same section of the Probability Calculator interface as the previous one, but with a teal arrow pointing from the 'Last Price' field towards the 'Interest Rate' field.

Interest Rate

The default risk-free interest rate is derived from last night's treasury market and is the rate that is equivalent to the option's expiration term. Users can change the interest rate by entering any other value.



This screenshot shows the bottom section of the Probability Calculator interface. It includes a 'Last Price' field, an 'Interest Rate' field, an 'Expiry' field with a dropdown menu, a 'Dividend Date' field with a date picker (YYYY-MM-DD), and a 'Time' field with a time picker (H, M, S). A teal arrow points from the 'Interest Rate' field towards the 'Expiry' field. On the far right, there are labels for 'Probabil' and 'Below Pri'.

Expiry

This will populate a drop down with all potential expiration dates. The default value will be the closest expiry date, or the option with the fewest days to expiration. Expiry date and time are customizable and modifications will display possible impact to an option's value.

Dividend Amount, Date & Frequency

Depending upon whether a stock pays a dividend or not, this section may or may not contain data. If there is data within the fields you can alter it by simply adding in the dividend amount and the corresponding ex. dividend date into the fields. The calculator will then price the options accordingly.

Time

The three boxes represent days, hours, and minutes left to expiry. This can be changed to reflect a non-standard expiry date or to perform "what if" analysis.

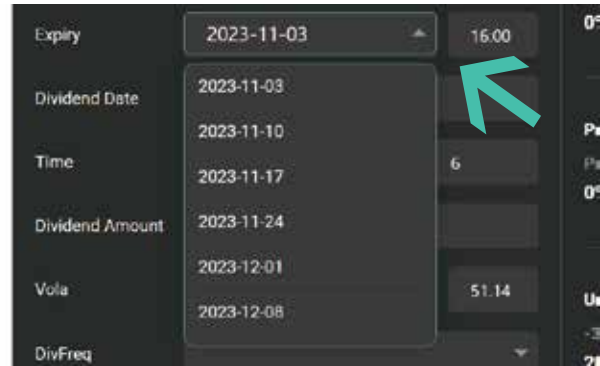
Volatility Level

This allows you to select a methodology with which to set the volatility level of the underlying. The possible selections include:

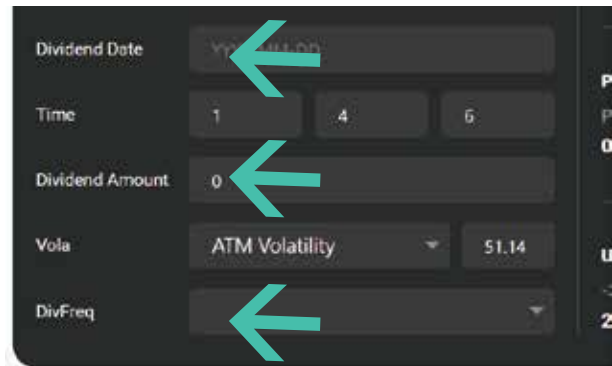
- The at-the-money volatility of the nearest term option
- The IV Index 30D (the implied volatility of the option with 30 days to expiry), IV index 60D, or IV Index 90D
- The HV 10D (the 10-day historical volatility of the underlying), HV 60D, HV 90D, HV 120D, HV 150D, or HV 180D

Calculations

When all inputs are set on the left side, input two prices into the P1 and P2 boxes and press confirm.



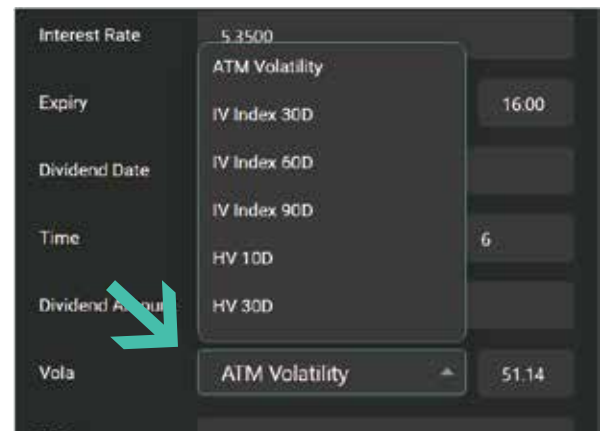
A screenshot of a software interface showing a dropdown menu for the 'Expiry' field. The dropdown is open, displaying a list of dates: 2023-11-03, 2023-11-10, 2023-11-17, 2023-11-24, 2023-12-01, and 2023-12-08. A green arrow points to the 2023-11-03 date. Other fields visible include Dividend Date, Time (6), Dividend Amount, Vola (51.14), and DivFreq.



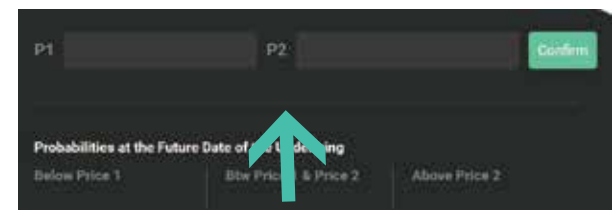
A screenshot of the 'Dividend' section of the interface. It shows fields for Dividend Date (YYYY-MM-DD), Time (1, 4, 6), Dividend Amount (0), Vola (ATM Volatility, 51.14), and DivFreq. Green arrows point to the Dividend Date, Dividend Amount, and DivFreq fields.



A screenshot of the 'Time' section of the interface. It shows fields for Dividend Date (YYYY-MM-DD), Time (1, 4, 6), and Dividend Amount (0). A green arrow points to the Time field.



A screenshot of the 'Volatility Level' dropdown menu. The dropdown is open, showing options: ATM Volatility, IV Index 30D, IV Index 60D, IV Index 90D, HV 10D, and HV 30D. A green arrow points to the IV Index 30D option. Other fields visible include Interest Rate (5.3500), Expiry, Dividend Date, Time (6), Dividend Amount, Vola (ATM Volatility, 51.14), and DivFreq.



A screenshot of the 'Calculations' section of the interface. It shows input fields for P1 and P2, a 'Confirm' button, and a section for 'Probabilities at the Future Date of Expiry' with sub-sections for 'Below Price 1', 'Btw Price 1 & Price 2', and 'Above Price 2'. A green arrow points to the 'Btw Price 1 & Price 2' section.

Calculations Continued...

With this data, the Probability Calculator takes volatility in options on the underlying stock, index or ETF, and factors in the current risk-free interest rate, plus any dividends. This sums up the key factors of what the market's activity is "implying" about future prices on the underlying. The Probability Calculator helps gauge the statistical likelihood of hitting certain price highs and lows through that time period.

The first section uses inputs to generate the probability of the following scenarios:

- Below Price 1
- Between Price 1 and Price 2
- Above Price 2

The next section expresses the statistical probability of touching the following:

- Price 1
- Price 2
- Both Price 1 and Price 2
- Either Price 1 or Price 2
- Neither Price 1 nor Price 2

The last section measures what the underlying prices look like for each standard deviation interval at the future date. In other words, the calculator will tell you that if a stock moves by X Standard Deviation, what that specific share price would be.

P1: 216 P2: 220 Confirm

Probabilities at the Future Date of the Underlying

Below Price 1	Btw Price 1 & Price 2	Above Price 2
35%	25%	41%

Probability of Touching

Price 1	Price 2	Both	Either	Neither
68.93%	81.87%	36.49%	85.38%	14.62%

Underlying Prices for each Standard Deviation Interval at the Future Date

-3	-2	-1	+1	+2	+3
200.17	206.1	212.2	224.95	231.6	238.46

Disclaimer

The Probability Calculator and 20-minute delayed options quotes are provided by IVolatility, and NOT BY OCC. The tool is being provided for educational purposes only. OCC makes no representation as to the timeliness, accuracy or validity of the information and this information should not be construed as a recommendation to purchase or sell a security, or to provide investment advice. For more information or help, please contact us at options@theocc.com.